Module 1: Proposed exercise

The solution of the exercises proposed in this document will not be given during class.

Exercise 1) [7 points]

Consider process composed by a tank of area *S = 2m²* and a heating lamp controlled by a tension $V_l(t)$. The water temperature (T) is given by the following:

$$\frac{dT(t)}{dt} = -0.5(q_l(t) - S \cdot T(t))$$

Where $q_l(t) = 10V^{-2}(t)$ is the heat produced by the lamp.

We are interested in the temperature $T(t)$ of the tank.

1. Determine the dynamic equation that define the system.
2. Classify the system.
3. Find and equilibrium point such that the tank temperature is 20 °C.
4. If the equilibrium point exists, determine, if possible, its stability.

Exercise 2) [5 points]

Given the following LTI System

$$\begin{cases} x_1(t) = x_1(t) - 3x_2(t) + u_1(t) \\ x_2(t) = x_1(t) + x_2(t) + u_2(t) \\ \\ y(t) = x_1(t) + u_1(t) \end{cases}$$

1. Define the transfer function.
2. Is the transfer function a solid approximation of the system? Motivate the answer.

Exercise 3) [7 points]

Consider the transfer function:

$$G(s) = \frac{100}{s^2 + 101s + 100}$$

and recalling the following:

1st Order Strictly Proper				*1st Order Proper*	*2nd Order: complex pole*	
T_s	T_r	T_{a5}	T_{a1}	$T_{a\varepsilon}$	$S\%$	$T_{a\varepsilon}$
$\approx 2.2T$	$\approx 0.7T$	$\approx 3T$	$\approx 4.7T$	$T \ln\frac{\lvert 1-\alpha\rvert}{0.01\varepsilon}$	$100e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}$	$-\frac{\ln 0.01\varepsilon}{\xi\omega_n}$

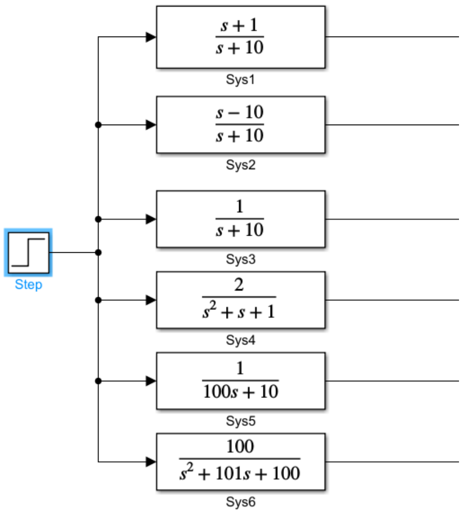
determine the following indexes for the response to a 10 unit step

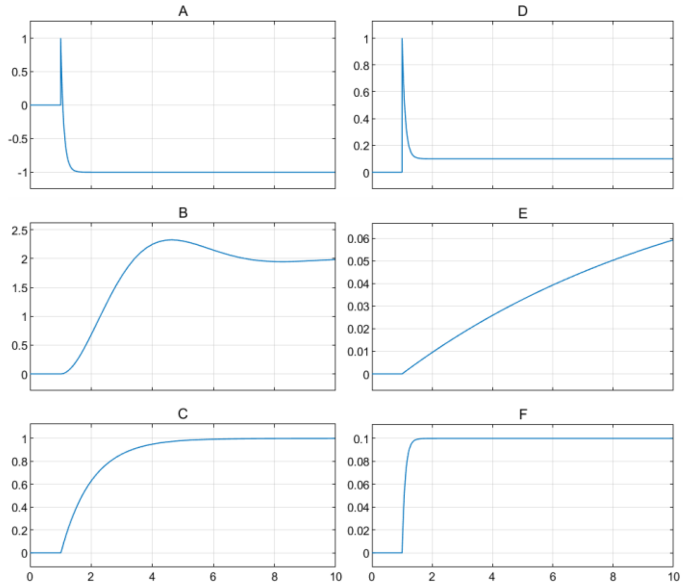
- Equilibrium value
- Settling time at 5%

- Overshooting percentage

Exercise 4) [6 point]

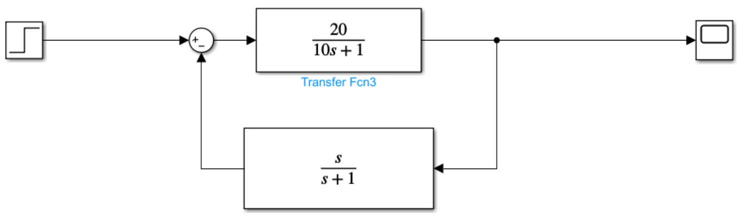
Label each step response with the correct system





Exercise 5) [7points]

Given the system



1. evaluate the response to the following signal:
   a. $\cos(100t) + \cos(t/100)$

2. Define the Bode Diagram

Consider process composed by a tank of area $S = 2m^2$ and a heating lamp controlled by a tension $V_l(t)$. The water temperature (T) is given by the following:

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Where $q_l(t) = 10V^{-2}(t)$ is the heat produced by the lamp.

We are interested in the temperature $T(t)$ of the tank.

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$S = 2m^2$

$$\frac{dT(t)}{dt} = -0.5(q_l(t) - ST(t)) \qquad q_l(t) = 10V^{-2}(t)$$

$$\dot{x}(t) = -0.5\left(\frac{10}{w^2(t)} - 2x(t)\right)$$

- LINEAR ?
- TIME INVARIANT ✓
- DYNAMIC ✓
- SISO ✓
- ~~PROPER~~ STRICTLY PROPER
- 1 ORDER ✓

EQUILIBRIUM S.T. $T = 20°C$

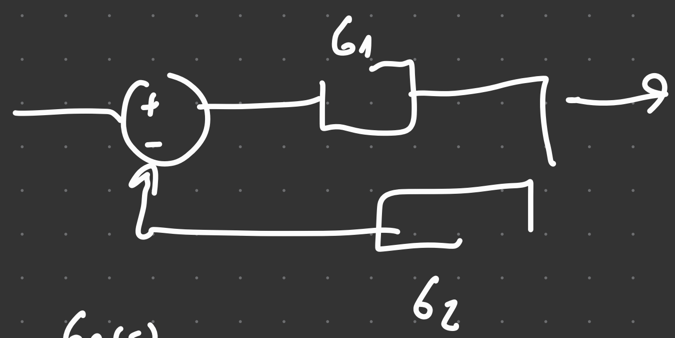
$(\bar{x}, \bar{w})$:

$$0 = -0.5\left(\frac{10}{w^2(t)} - 2 \cdot 20\right) \longrightarrow \frac{10}{w^2(t)} = 40 \; ; \; w^2(t) = 1/4 \longrightarrow w(t) = 1/2$$

$(20, 0.5)$

STABILITY

$$\begin{cases} \delta\dot{x}(t) = A\delta x(t) + B\delta w(t) \\ \delta y(t) = C\delta x(t) + D\delta w(t) \end{cases} \qquad A = [\qquad ?$$



$$\frac{G_A(s)}{1 + G_A(G_B)}$$